

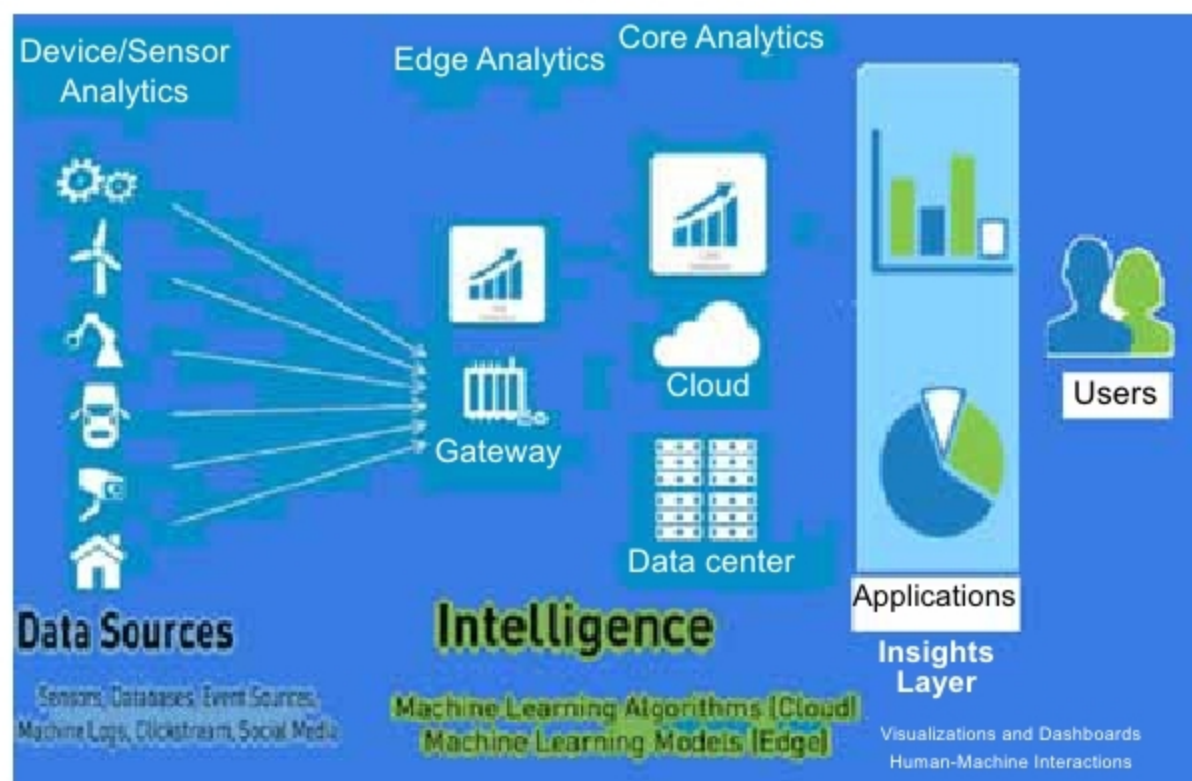


Figure 2: Edge computing 3-tier architecture

flowing in both directions, and prevents unauthorised control of the devices from the outside world. It also facilitates the devices' life cycle management, the phases of which are shown in Figure 3.

The IoT solution consists of Web application interfaces such as Cockpit that provide an interface to administer and configure the devices for management and control. Cockpit can be implemented using front-end technologies like Angular and Node.js.

IoT back-end systems have various components for data ingestion (that use supportive frameworks based on the technology stack, such as Apache Camel and Spark Streaming), for orchestration and processing (that define the business logic and workflows like NiFi and Apache Spark), for modelling and analytics (examples include R, Python, Spark ML, Kibana, Elastic Search and TensorFlow), and for data storage (examples include MongoDB, MySQL, PostgreSQL, Redis, CouchDB and Cassandra).

Major cloud service providers like Azure and AWS have a rich set of technologies and services for aiding IoT, both in the cloud and

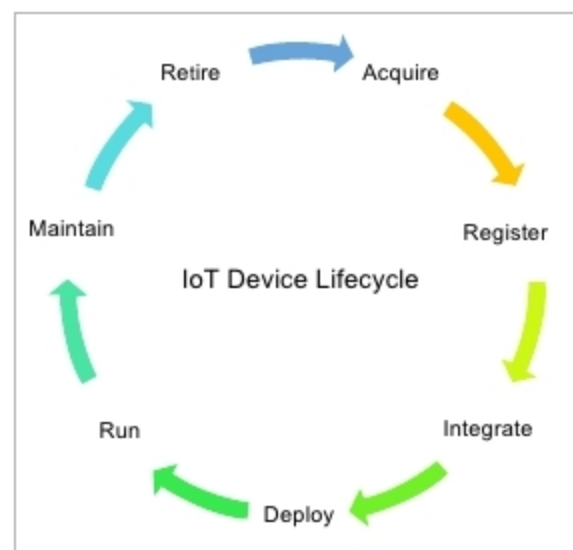


Figure 3: Life cycle of an IoT device

in the edge. Azure IoT Hub, Azure API Management, Azure IoT Edge, AWS IoT and AWS API Gateway are a few examples.

A few open source technology tools, frameworks and projects for IoT systems are LFEEdge, EdgeX Foundry, KubeEdge, HomeEdge, Kura, Kapua, Spring Boot, Eclipse Hono, Kaa, OpenIoT, DeviceHive, OpenThread, etc, to name a few. Roof Computing is a standard from IEEE to drive edge computing standardisation.

Open source security and IoT

More connections mean more pathways and back doors that can be exploited by hackers. Open source components are vulnerable to attacks as the developers of these components are not security professionals. In order to use open source for IoT on the edge, the software needs to be analysed end-to-end because the code on the device is no longer standalone. Once scanned, all vulnerabilities should be addressed to create robust systems.

Edge computing has enriched the IoT world by giving the users of smart devices a complete sense of control and security, thus improving the confidentiality, integrity and availability of IoT data. Edge can be compared to the hybrid cloud, which is very popular and is being increasingly adopted by enterprises moving their businesses to cloud. Edge services and infrastructure, too, will see increased demand in the coming years, not only in the IoT domain but also in other emerging markets. **END** 🐧

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